# MAGWADI ADELAIDE PALANE

Tzaneen, Limpopo, South Africa • +27 78 632 5143 • magwadiadelaide28@gmail.com • LinkedIn Profile • GitHub Portfolio

## PROFESSIONAL SUMMARY

Results-driven **Information and Communication Technology (BICT) Graduate** with practical experience in solutions engineering, data modeling, API validation pipelines, and business intelligence reporting. Proven track record in orchestrating event-driven webhook oracles, querying GraphQL endpoints, enforcing strict data integrity (SQL/JSON Schema), and translating complex datasets into actionable stakeholder dashboards. Combines solid academic foundations in Data Scalability, Analytics, and Database Systems with hands-on enterprise integration skills and POPIA compliance awareness to deliver reliable technical solutions.

## CORE COMPETENCIES & TECHNICAL TOOLKIT

- **Solutions Engineering & Integrations:** REST APIs, Webhooks, GraphQL (Blocksync), JSON Schema (Draft-07), Workflow Automation (Make.com, Zapier), Data Mapping.
- **Data Analytics & BI:** SQL (PostgreSQL, Supabase), Advanced Excel (Power Query, PivotTables, Formulas), Power BI, Python, Data Cleaning & Scalability.
- **Enterprise Systems & Modeling:** CRM Data Modeling, Record-Triggered Flows, SLA Monitoring, Human-in-the-Loop (HITL) Exception Handling, Airtable Relational Bases.
- **Governance & Compliance:** POPIA Guidelines, Data Confidentiality, KYC Verification Support, Verifiable Claim Provenance.

## TECHNICAL PROJECTS & PORTFOLIO

**Technical Solutions & Platform Engineering Portfolio** **github.com/Adelaide2828/solutions-engineer-portfolio**

**Event-Driven Webhook Validation Oracle & Data Firewall *(Make.com, Supabase SQL, Airtable, Postman)***

- Architected an event-driven validation pipeline inspecting 100% of incoming payloads against strict JSON Schema contracts before database ingestion.
- Configured conditional routing logic quarantining anomalous records (HTTP 422), dispatching real-time alerts to ops, and committing clean records (HTTP 200).
- Authored automated Postman API test collections to perform regression testing across multi-scenario payloads.

**Decentralized Provenance & Analytical Dashboard *(NumiKnow / IXO Protocol, GraphQL, Modern UI)***

- Engineered an interactive web analytics dashboard querying Blocksync GraphQL to track verified educational outcomes across 5 regional school domains.
- Established a 4-Tier Data Trust Framework (Cryptographic Proofs, Device Telemetry, Curated Datasets, Assertions), reducing audit cycles from 3 weeks to real-time.

**Agentic Verification Oracle & Human-in-the-Loop Governance *(Python, Policy Manifests, Cosmos Authz)***

- Implemented a deterministic verification engine enforcing declarative Policy Manifest Rubrics across facilitator credentials, geofenced GPS, and score outliers.
- Automated 85% of standard grant claim verifications with sub-second execution while systematically escalating complex edge cases to human auditors.

## PROFESSIONAL EXPERIENCE

**IXO Solutions Engineering Cohort Programme — *Solutions Engineer Trainee*** **2025 – 2026**

- Supported project partners in translating operational and grant funding requirements into technical integration blueprints and POD system definitions.
- Monitored real-time submission pipelines across pending, approved, and rejected states using GraphQL queries and event logs.
- Assisted participants and partner organizations in completing KYC verification workflows adhering to strict data verification protocols.
- Collaborated across technical cohort pods to troubleshoot validation bottlenecks and optimize claim processing turnaround.

**Community Educational Initiative | Youth Development Project — *Data Administrator & Project Coordinator*** **Jan 2026 – Jul 2026**

- Administered educational datasets tracking 100+ Grade 6–7 learners' attendance, test metrics, and performance progressions across Hoedspruit Hub initiatives.
- Cleaned, transformed, and standardized messy field spreadsheets using Advanced Excel and Power Query, removing duplicates and correcting formatting inconsistencies.
- Maintained strict POPIA compliance by implementing anonymization and data masking protocols on all student and facilitator identifying records.
- Prepared weekly and monthly analytical reports with visual charts for program managers, identifying trends in student numeracy improvement.

## EDUCATION

**Bachelor of Information and Communication Technology (BICT) — *NQF Level 7*** **Conferred May 2025**

*University of Mpumalanga* | Completed: December 2024

- **Data Analytics & Statistics:** Data Analytics 321, Data Scalability & Analytics 202, Statistics for ICT 201, Mathematics for Computing.
- **Databases & Architecture:** Database Systems 201, Introduction to Databases 101, Data Structures & Algorithms 202.
- **Software & Systems:** System Development Project 300, Principles of Software Engineering 201, Programming Techniques (I, II, III), Web Development.
- **Infrastructure & Networks:** Introduction to IoT 202, Communication Networks (101, 102, 322), Cybersecurity (202, 312), Operating Systems.

## CERTIFICATES, RECOGNITION & LANGUAGES

- **Certificate of Appreciation – Program Management:** Hoedspruit Hub (Awarded for outstanding operational management and data tracking of Grade 7 Enrichment Program) | *Sept 2026*
- **Languages:** English (Professional Working Proficiency / Fluent), Sepedi (Native / Fluent), Zulu (Conversational).